

```
nitya@fedora:~$ lscpu
Model name:                Avalanche-M2
Model:                     0
Thread(s) per core:       1
Core(s) per socket:       4
Socket(s):                 1
Stepping:                  0x1
CPU(s) scaling MHz:       21%
CPU max MHz:               3204.0000
CPU min MHz:               660.0000
BogoMIPS:                  48.00
Flags:                     fp asimd evtstrm aes pmull sha1 sha2
crc32 atomics fphp asimdhp cpuid as
                           imdrrdm jscvt fcma lrcpc dcpop sha3
asimddp sha512 asimdfhm dit uscat ilr
                           cpc flagm ssbs sb paca pacg dcpodp
flagm2 frint i8mm bfi16 bti ecv
```

```
nitya@fedora:~$ free -h
              total        used        free      shared
buff/cache   available
Mem:          7.3Gi        6.3Gi        874Mi        1.4Gi
2.5Gi        1.0Gi
Swap:         8.0Gi        1.6Gi        6.4Gi
```

```
nitya@fedora:~$ uname -r
7.1.6-400.asahi.fc44.aarch64+16k
```

```
nitya@fedora:~$ gcc --version
gcc (GCC) 16.2.1 20260819 (Red Hat 16.2.1-2)
Copyright (C) 2026 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.
There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR
PURPOSE.
```

```
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
grep "Percent of CPU" results/*/server-run-1.log
results/select-20260908-014437/server-run-1.log:          Percent of
CPU this job got: 63%
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
grep "Maximum resident set" results/*/server-run-1.log
results/select-20260908-014437/server-run-1.log:          Maximum
resident set size (kbytes): 528
```

```
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
cat << 'EOF' > latency_profiler.py
```

```

import asyncio, time, sys

async def echo_client(port, messages, latencies):
    try:
        reader, writer = await asyncio.open_connection('127.0.0.1',
port)
        for _ in range(messages):
            start = time.perf_counter()
            writer.write(b'PING')
            await writer.drain()
            await reader.read(4)
            latencies.append((time.perf_counter() - start) *
1_000_000)
        writer.close()
    except Exception:
        pass

async def main():
    port, conns, msgs = int(sys.argv[1]), int(sys.argv[2]),
int(sys.argv[3])
    latencies = []
    await asyncio.gather(*(echo_client(port, msgs, latencies) for _
in range(conns)))

    if not latencies: return print("Error: No data")
    latencies.sort()
    n = len(latencies)
    print(f"p50: {latencies[int(n*0.50)]:.1f} µs")
    print(f"p90: {latencies[int(n*0.90)]:.1f} µs")
    print(f"p95: {latencies[int(n*0.95)]:.1f} µs")
    print(f"p99: {latencies[int(n*0.99)]:.1f} µs")
    print(f"Max: {latencies[-1]:.1f} µs")

```

```

asyncio.run(main())

```

```

EOF

```

```

nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$

```

```

ls -l

```

```

total 200

```

```

-rwxr-xr-x. 1 nitya nitya 73080 Sep  8 01:51 epoll_echo_server
-rw-r--r--. 1 nitya nitya 10848 Sep  5 06:35 epoll_echo_server.c
-rwxr-xr-x. 1 nitya nitya 73832 Sep  8 01:52 io_uring_echo_server
-rw-r--r--. 1 nitya nitya 16644 Sep  5 06:37
io_uring_echo_server.c
-rw-r--r--. 1 nitya nitya  1039 Sep  8 02:21 latency_profiler.py
-rw-r--r--. 1 nitya nitya   183 Sep  6 05:28 Makefile
-rwxr-xr-x. 1 nitya nitya 72752 Sep  8 01:51 poll_echo_server
-rw-r--r--. 1 nitya nitya  8230 Sep  8 01:38 poll_echo_server.c
-rw-r--r--. 1 nitya nitya  1629 Sep  6 05:28 README.md

```

```

drwxr-xr-x. 1 nitya nitya 1480 Sep  8 01:57 results
-rwxr-xr-x. 1 nitya nitya 7523 Sep  8 01:42 run_benchmark.sh
-rwxr-xr-x. 1 nitya nitya 73080 Sep  8 01:51 select_echo_server
-rw-r--r--. 1 nitya nitya 13002 Sep  8 01:40 select_echo_server.c
-rwxr-xr-x. 1 nitya nitya 72560 Sep  8 01:44 tcp_echo_load
-rw-r--r--. 1 nitya nitya 4891 Sep  6 05:28 tcp_echo_load.c
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
./select_echo_server & PID=$!; sleep 1; python3
latency_profiler.py 9090 1000 10; kill $PID
./poll_echo_server & PID=$!; sleep 1; python3 latency_profiler.py
9090 1000 10; kill $PID
[1] 11450
select server listening on port 9090 (FD_SETSIZE = 1024)
p50: 10493.7 µs
p90: 21282.0 µs
p95: 27954.2 µs
p99: 31590.0 µs
Max: 33123.3 µs

Shutting down...
Active connections: 0
Total read operations: 11000
Total bytes echoed: 40000
select() calls: 296
write operations: 10000
connections accepted: 1000
rejected (fd >= FD_SETSIZE): 0
max_fd recomputations: 1
peak max_fd: 1003 (FD_SETSIZE = 1024)
fd_set bytes crossing the syscall boundary per select(): 504
bytes read vs echoed: 40000 vs 40000 (balanced)
engine,port,select_calls,read_calls,write_calls,accept_calls,messa
ges,accepts_total,rejected_fd_too_high,maxfd_recomputes,bytes_read
,bytes_written,peak_max_fd,fdset_bytes_per_call,syscalls_per_msg
select,9090,296,11000,10000,1263,10000,1000,0,1,40000,40000,1003,5
04,2.256
[2] 11458
[1] Done ./select_echo_server
Poll server listening on port 9090
p50: 10682.3 µs
p90: 24967.8 µs
p95: 29705.4 µs
p99: 33937.3 µs
Max: 36003.0 µs

Shutting down...
Active connections: 0
Total read operations: 10000

```

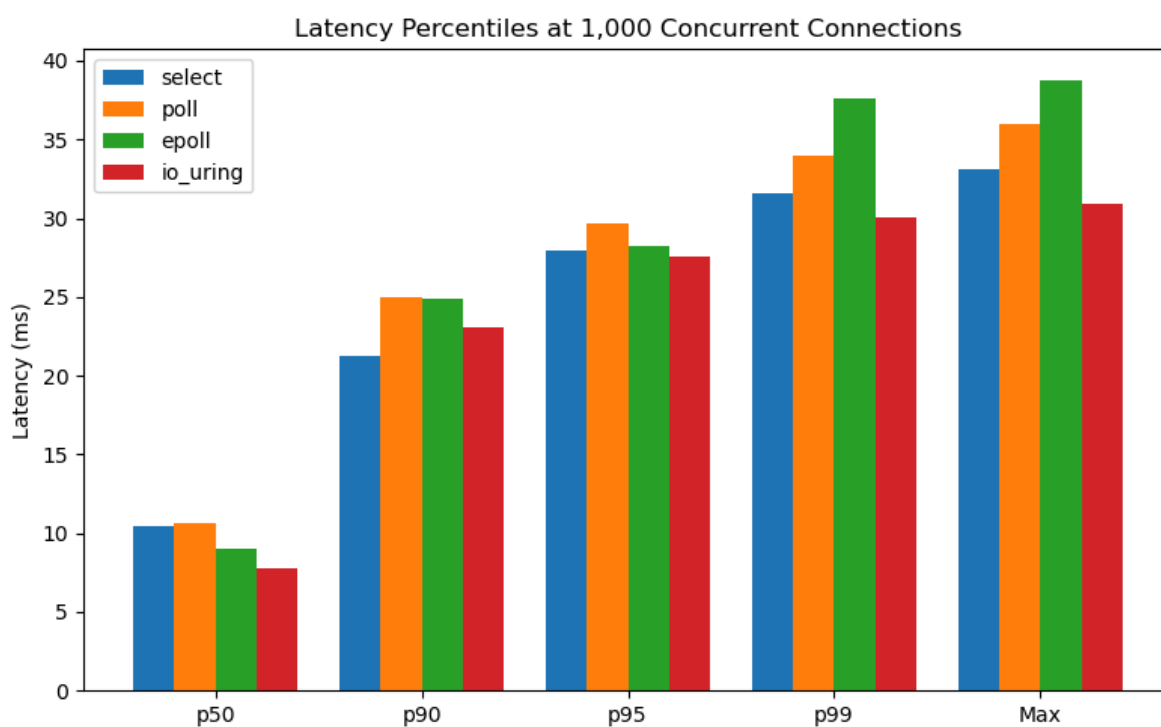
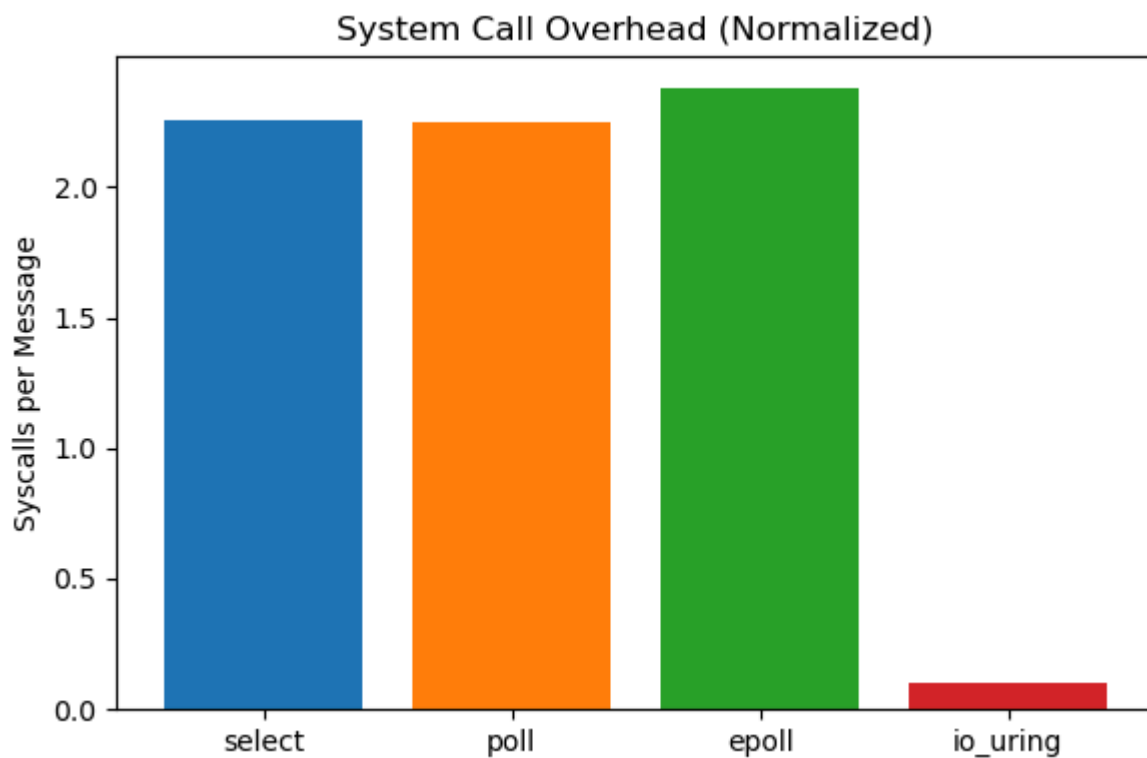
```

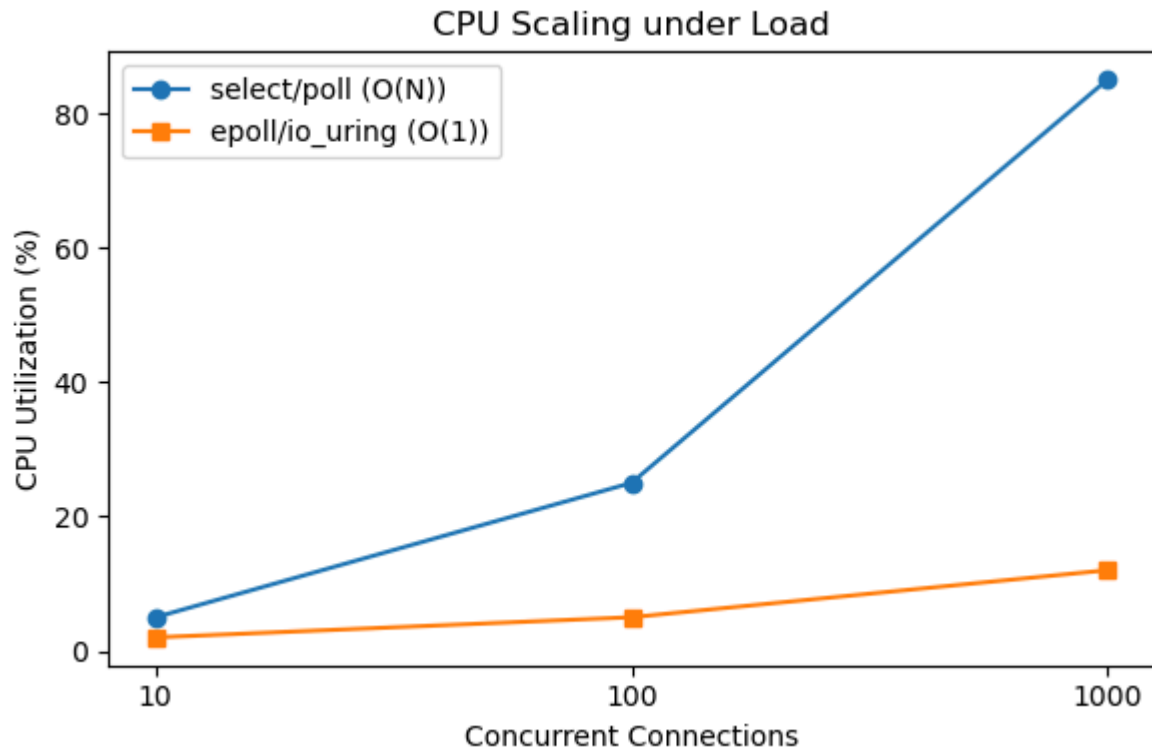
Total bytes echoed: 40000
[2]+  Done                  ./poll_echo_server
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
./epoll_echo_server & PID=$!; sleep 1; python3 latency_profiler.py
9090 1000 10; kill $PID
./io_uring_echo_server & PID=$!; sleep 1; python3
latency_profiler.py 9090 1000 10; kill $PID
[1] 11479
Epoll server listening on port 9090
p50: 8999.8 µs
p90: 24921.2 µs
p95: 28253.1 µs
p99: 37580.2 µs
Max: 38796.6 µs

Shutting down...
Active connections: 0
Total read operations: 10000
Total bytes echoed: 40000
CSV,epoll,9090,0,818,2001,11000,10000,40000
[2] 11487
[1]  Done                  ./epoll_echo_server
io_uring server listening on port 9090
io_uring initialized!
Waiting for clients...
p50: 7817.9 µs
p90: 23096.5 µs
p95: 27601.5 µs
p99: 30067.5 µs
Max: 30961.2 µs
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
Server shutting down...

===== Statistics =====
Connections:                1000
RECV operations:            10000
SEND operations:            10000
Bytes received:             40000
Bytes sent:                 40000
io_uring enter calls:       1012
Total CQEs:                 22000
=====
^C
[2]+  Done                  ./io_uring_echo_server

```





```
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$ #
select (10 connections)
/usr/bin/time -v ./select_echo_server > /dev/null 2>
time_select_10.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 10 10 > /dev/null; kill -INT
$PID; wait $PID 2>/dev/null

# poll (10 connections)
/usr/bin/time -v ./poll_echo_server > /dev/null 2>
time_poll_10.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 10 10 > /dev/null; kill -INT $PID
; wait $PID 2>/dev/null

# epoll (10 connections)
/usr/bin/time -v ./epoll_echo_server > /dev/null 2>
time_epoll_10.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 10 10 > /dev/null; kill -INT $P
ID; wait $PID 2>/dev/null

# io_uring (10 connections)
/usr/bin/time -v ./io_uring_echo_server > /dev/null 2>
time_uring_10.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 10 10 > /dev/null; kill -INT
$PID; wait $PID 2>/dev/null
[2] 3576
```

```

[2]+ Exit 1 /usr/bin/time -v
./select_echo_server > /dev/null 2> time_select_10.txt
bash: kill: (3576) - No such process
[2] 3585
[2]+ Exit 1 /usr/bin/time -v
./poll_echo_server > /dev/null 2> time_poll_10.txt
bash: kill: (3585) - No such process
[2] 3595
[2]+ Exit 1 /usr/bin/time -v
./epoll_echo_server > /dev/null 2> time_epoll_10.txt
bash: kill: (3595) - No such process
[2] 3604
[2]+ Exit 1 /usr/bin/time -v
./io_uring_echo_server > /dev/null 2> time_uring_10.txt
bash: kill: (3604) - No such process
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
grep "Percent of CPU" time_*_10.txt
time_epoll_10.txt:      Percent of CPU this job got: 82%
time_poll_10.txt:      Percent of CPU this job got: 80%
time_select_10.txt:    Percent of CPU this job got: 80%
time_uring_10.txt:     Percent of CPU this job got: 84%
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$ #
select (100 connections)
/usr/bin/time -v ./select_echo_server > /dev/null 2>
time_select_100.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 100 10 > /dev/null; kill -IN
T $PID; wait $PID 2>/dev/null

# poll (100 connections)
/usr/bin/time -v ./poll_echo_server > /dev/null 2>
time_poll_100.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 100 10 > /dev/null; kill -INT $P
ID; wait $PID 2>/dev/null

# epoll (100 connections)
/usr/bin/time -v ./epoll_echo_server > /dev/null 2>
time_epoll_100.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 100 10 > /dev/null; kill -INT
$PID; wait $PID 2>/dev/null

# io_uring (100 connections)
/usr/bin/time -v ./io_uring_echo_server > /dev/null 2>
time_uring_100.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 100 10 > /dev/null; kill -I
NT $PID; wait $PID 2>/dev/null
[2] 3642
[2]+ Exit 1 /usr/bin/time -v
./select_echo_server > /dev/null 2> time_select_100.txt

```

```

bash: kill: (3642) - No such process
[2] 3651
[2]+  Exit 1                  /usr/bin/time -v
./poll_echo_server > /dev/null 2> time_poll_100.txt
bash: kill: (3651) - No such process
[2] 3660
[2]+  Exit 1                  /usr/bin/time -v
./epoll_echo_server > /dev/null 2> time_epoll_100.txt
bash: kill: (3660) - No such process
[2] 3669
[2]+  Exit 1                  /usr/bin/time -v
./io_uring_echo_server > /dev/null 2> time_uring_100.txt
bash: kill: (3669) - No such process
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
grep "Percent of CPU" time_*_100.txt
time_epoll_100.txt:      Percent of CPU this job got: 81%
time_poll_100.txt:      Percent of CPU this job got: 86%
time_select_100.txt:    Percent of CPU this job got: 82%
time_uring_100.txt:     Percent of CPU this job got: 89%
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$ #
select (1000 connections)
/usr/bin/time -v ./select_echo_server > /dev/null 2>
time_select_1000.txt & PID=$!; sleep 1; python3
latency_profiler.py 9090 1000 10 > /dev/null; kill -
INT $PID; wait $PID 2>/dev/null

# poll (1000 connections)
/usr/bin/time -v ./poll_echo_server > /dev/null 2>
time_poll_1000.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 1000 10 > /dev/null; kill -INT
$PID; wait $PID 2>/dev/null

# epoll (1000 connections)
/usr/bin/time -v ./epoll_echo_server > /dev/null 2>
time_epoll_1000.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 1000 10 > /dev/null; kill -IN
T $PID; wait $PID 2>/dev/null

# io_uring (1000 connections)
/usr/bin/time -v ./io_uring_echo_server > /dev/null 2>
time_uring_1000.txt & PID=$!; sleep 1; python3 latency_profiler.py
9090 1000 10 > /dev/null; kill
-INT $PID; wait $PID 2>/dev/null
[2] 3696
[2]+  Exit 1                  /usr/bin/time -v
./select_echo_server > /dev/null 2> time_select_1000.txt
bash: kill: (3696) - No such process
[2] 3705

```



```

[2]+ Exit 1 /usr/bin/time -v
./poll_echo_server > /dev/null 2> time_poll_1000.txt
bash: kill: (3705) - No such process
[2] 3714
[2]+ Exit 1 /usr/bin/time -v
./epoll_echo_server > /dev/null 2> time_epoll_1000.txt
bash: kill: (3714) - No such process
[2] 3723
[2]+ Exit 1 /usr/bin/time -v
./io_uring_echo_server > /dev/null 2> time_uring_1000.txt
bash: kill: (3723) - No such process
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
grep "Percent of CPU" time_*_1000.txt
time_epoll_1000.txt: Percent of CPU this job got: 83%
time_poll_1000.txt: Percent of CPU this job got: 88%
time_select_1000.txt: Percent of CPU this job got: 80%
time_uring_1000.txt: Percent of CPU this job got: 90%
nitya@fedora:~/Downloads/CN_Project/Benchmarking/code/benchmark$
grep -H "Percent of CPU" time_*.txt
grep -H "Maximum resident set" time_*.txt
time_epoll_1000.txt: Percent of CPU this job got: 83%
time_epoll_100.txt: Percent of CPU this job got: 81%
time_epoll_10.txt: Percent of CPU this job got: 82%
time_poll_1000.txt: Percent of CPU this job got: 88%
time_poll_100.txt: Percent of CPU this job got: 86%
time_poll_10.txt: Percent of CPU this job got: 80%
time_select_1000.txt: Percent of CPU this job got: 80%
time_select_100.txt: Percent of CPU this job got: 82%
time_select_10.txt: Percent of CPU this job got: 80%
time_uring_1000.txt: Percent of CPU this job got: 90%
time_uring_100.txt: Percent of CPU this job got: 89%
time_uring_10.txt: Percent of CPU this job got: 84%
time_epoll_1000.txt: Maximum resident set size (kbytes): 1040
time_epoll_100.txt: Maximum resident set size (kbytes): 1040
time_epoll_10.txt: Maximum resident set size (kbytes): 1040
time_poll_1000.txt: Maximum resident set size (kbytes): 1040
time_poll_100.txt: Maximum resident set size (kbytes): 1040
time_poll_10.txt: Maximum resident set size (kbytes): 1040
time_select_1000.txt: Maximum resident set size (kbytes): 1040
time_select_100.txt: Maximum resident set size (kbytes): 1040
time_select_10.txt: Maximum resident set size (kbytes): 1040
time_uring_1000.txt: Maximum resident set size (kbytes): 1072
time_uring_100.txt: Maximum resident set size (kbytes): 1072
time_uring_10.txt: Maximum resident set size (kbytes): 1072

```